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Striatoids A−F, Cyathane Diterpenoids with Neurotrophic Activity from Cultures of the Fungus *Cyathus striatus*

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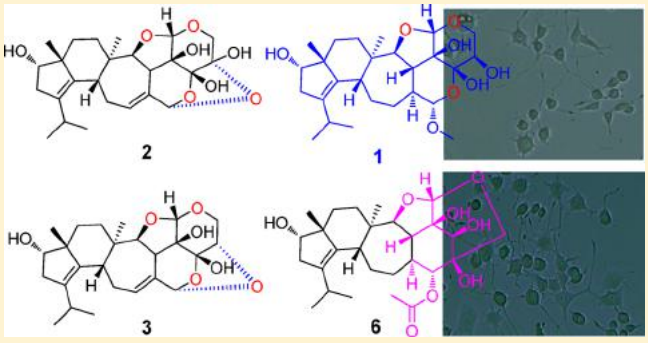
Supporting Information

ABSTRACT: Six new highly oxygenated polycyclic cyathane-xylosides, named striatoids A−F (**1**−**6**), were isolated from the cultures of the basidiomycete *Cyathus striatus*. Their structures were established by comprehensive spectroscopic analysis including 2D NMR (HMBC, HSQC, ROESY, ^{1}H−^{1}H−COSY) and HRESIMS experiments. Compounds **2** and **3** possess an unusual 15,4′-ether ring system. The isolated compounds dose-dependently enhanced nerve growth factor (NGF)-mediated neurite outgrowth in rat pheochromocytoma (PC-12) cells.



Nerve growth factor (NGF) is the first reported and best characterized neurotrophic factor that is recognized as an important regulatory substance in the nervous system.[1] Thus, NGF is expected to have therapeutic efficacy for the treatment of neurodegenerative diseases such as Alzheimer's disease and Parkinson's disease. However, it cannot cross the brain−blood barrier because of it is a high molecular weight polypeptide and is easily metabolized by peptidases under physiological conditions.[1] To address this issue, considerable efforts have been made to find small molecules that have neurotrophic properties and/or that are capable of enhancing the action of endogenous neurotrophic factors.

Basidiomycetes are known to produce a broad spectrum of secondary metabolites. A number of cyathane diterpenoids with an unusual 5/6/7 tricyclic skeleton, including their xylosides, were isolated from a diverse variety of higher Basidiomycetes of the genera *Cyathus, Hericium, and Sarcodon*.[2] These diterpenes have been proven to display a wide range of biological properties, including anti-inflammatory,[3] antimicrobial,[4] anti-tumor,[5] and antagonism toward the kappa-opioid receptor.[6] Several of the erinacine (e.g., erinacines B, E) and the scabronine (e.g., scabronines G, A) natural products were found to stimulate the synthesis of NGF in human nerve cells,[7] which indicates their potential as therapeutic agents to treat neurodegenerative ailments.

The fungus *Cyathus striatus* (Huds. ex Pers.) Willd, belonging to the family Nidulariaceae, is known as the bird's nest fungi. Previous chemical studies on *C. striatus* resulted in the discovery of antitumor cyathane-type metabolites such as the striatals A, B, C, and D.[5] In the course of our ongoing search for biologically active substances from higher Basidiomycetes,[8] chemical investigations of cultures of *C. striatus* were carried out, leading to the isolation of six new cyathane diterpenoids, striatoids A−F (**1**−**6**). The striatoids B (**2**) and C (**3**) possessed a rare 15,4′-ether ring system. In this paper, we report the isolation and structure determination of **1**−**6** and their enhancement of NGF-induced neutrite outgrowth in PC-12 cells.

■ RESULTS AND DISCUSSION

The organic extract obtained from the culture filtrates of *C. striatus* was fractionated as reported in detail in the Experimental Section, yielding six new cyathane-xylosides, striatoids A−F (**1**−**6**, Figure 1).

Compound **1** was isolated as a white powder. The molecular formula of **1** was determined to be $C_{26}H_{40}O_8$, with seven degrees of unsaturation, based on the HRESIMS at *m/z* 503.2603 [M + Na]$^+$. The ^{13}C NMR data (Table 1) revealed 26 carbon resonances, corresponding to five methyls (including one methoxy), six methylenes (including an oxygenated one), nine methines (including three oxygenated and two ketal carbons with typical signals at δ_C 103.0 (C-15) and 105.8 (C-1′)), and six quaternary carbons (including two olefinic, an oxygenated, and a ketal carbon). Overall consideration of the NMR data of compound **1** suggested a similarity to striatin A.[9] Detailed analysis of the 1D NMR together with the HMBC and COSY data revealed the main differences between compound **1** and striatin A were the absence of a double bond between C-11 and C-12 and an additional hydroxy group at C-1, which were supported by the ^{1}H−^{1}H COSY correlations of H-1/H-2 and H-10/H-11/H-12/H-13/H-14 and the HMBC correlations from H-17 to C-1, C-8, C-9, H-16 to C-5, C-6, C-7, C-14, and

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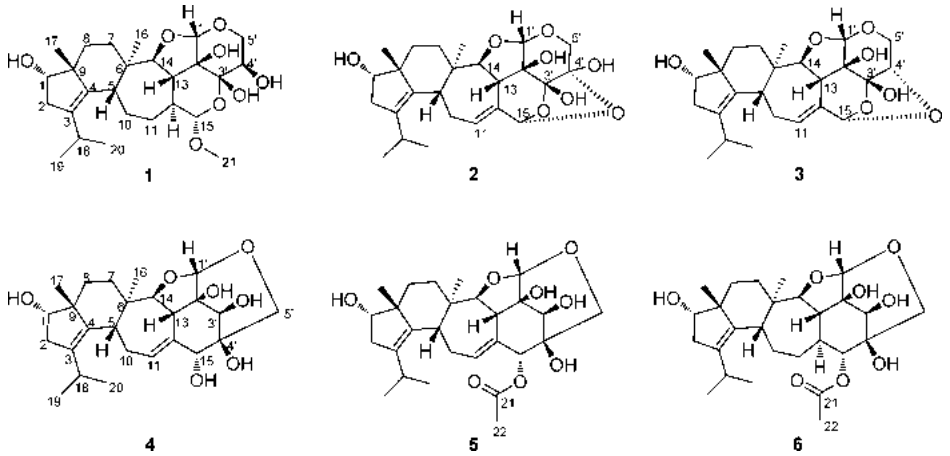

Figure 1. Structures of compounds **1**−**6**.

H-15 to C-12 (Figure 2). Hence, the planar structure of **1** was elucidated as shown.

The stereochemistry assigment of **1** was achieved by NOE effects along with biogenetic considerations. In the ROESY spectrum in DMSO-d_6, the correlations of H-1/17-Me, 17-Me/H-5/H-13/H-15, and H-1′/H-13/2′-OH/3′-OH/5′β-H suggested that the 1-OH and 15-methoxy groups were α-oriented and that 2′-OH and 3′-OH were β-oriented (Figure 3). 5′β-H showed a triplet signal with a J value of 10.8 Hz, revealing that 4′-H was α-oriented. Thus, the structure of **1** was determined and named striatoid A.

Compound **2** was obtained as a white powder. Its molecular formula was deduced as $C_{25}H_{34}O_8$ from the HRESIMS at m/z 485.2139 $[M + Na]^+$, indicating nine degrees of unsaturation. Its ^{13}C NMR spectrum (Table 1) displayed 25 carbon signals, suggesting that compound **2** shared the same cyathane-type skeleton as **1**. The main differences between the two compounds were the absence of the methoxyl at C-15, an oxygenated methine at δ_C 69.0 (C-4′) in **1** replaced by a ketal carbon at δ_C 93.2 (C-4′) in **2**, and the presence of a trisubstituted double bond (δ_C 124.9 (C-11), δ_H 5.43, 1H, m; δ_C 137.1 (C-12)). In the $^1H-^1H$ COSY spectrum, the correlations of H-5/H-10/H-11 indicated that the double bond was located at C-11 and C-12, which was supported by the HMBC correlations from H-13, H-14, and H-15 to C-12. In the HMBC spectrum, the correlations from H-15 to both ketal carbons C-3′ (δ_C 100.6) and C-4′ (δ_C 93.2) suggested that a C−O−C bond was formed between C-15 and C-4′, constructing a new ring system, which was consistent with the degrees of unsaturation. Detailed analysis of the 2D NMR (HMBC and ROESY) data revealed the remaining parts of **2** were the same as those of **1** including the stereoconfigurations. Thus, the structure of **2** was determined as striatoid B.

Compound **3** had the molecular formula $C_{25}H_{34}O_7$, as determined by its HRESIMS at m/z 447.2367 ($[M + H]^+$), 16 mass units less than that of **2**. The ^1H and ^{13}C NMR spectroscopic data of **3** (Table 1) were quite similar to those of **2**, and the main differences between **2** and **3** were an oxygenated methine replacing a ketal carbon, confirmed by the $^1H-^1H$ COSY correlations of H-4′ (δ_H 3.53, brs) to H-5′ (δ_H 3.67, d, J = 13.6 Hz; δ_H 3.99, d, J = 13.6 Hz) and the HMBC correlations of δ_H 5.64 (1H, s, H-15) to δ_C 74.2 (C-4′). Analysis of the spectroscopic data established the structure of **3** as striatoid C.

Compound **4** was obtained as a white powder, having the molecular formula $C_{25}H_{36}O_7$ based on the HRESIMS at m/z 471.2337 $[M + Na]^+$. The ^{13}C NMR spectrum (Table 2) displayed 25 signals for four methyls (two singlets and two doublets), five methylenes (one oxygenated), nine methines (one olefinic, two ketal, and four oxygenated), and seven quaternary carbons (two oxygenated and three olefinic). The ^1H and ^{13}C NMR data of **4** were quite similar to those of erinacine E, which had been isolated from the mycelia of *Hericium erinaceum*,[10] with the difference of a methylene of erinacine E being an oxygenated methine at δ_C 77.0 (C-1) in **4**. The difference was further supported by the HMBC correlations of δ_H 3.72 (1H, m, H-1) to δ_C 37.7 (C-2) and 52.0 (C-9). In the ROESY spectrum (Figures 4), the correlations of H-1/H-17 suggested that 1-OH was α-oriented, and the remainder of **4** was the same as erinacine E. Thus, compound **4** was assigned as striatoid D.

Compound **5** had the molecular formula of $C_{27}H_{38}O_8$ on the basis of HRESIMS at m/z 513.2444 $[M + Na]^+$, 42 mass units higher than that of **4**. The 1D NMR spectroscopic data of **5** (Table 2) were similar to those of **4**, and the main difference was an additional acetyl group (δ_C 170.6 (C-21); δ_C 21.1 (C-22), δ_H 2.07, 3H, s). The HMBC correlation of δ_H 5.86 (1H, s, H-15) to δ_C 170.6 (C-21) confirmed the location of the acetyl group at C-15. Detailed analysis of the 2D NMR (HMBC, COSY, COSY, and ROESY) spectroscopic data established the structure of **5** as striatoid E.

Compound **6** was isolated as a white powder. Its molecular formula was deduced as $C_{27}H_{40}O_8$ on the basis of the HRESIMS at m/z 515.2598 $[M + Na]^+$, two mass units higher than that of **5**. Detailed analysis of the 1D (Table 2) and 2D NMR data suggested that the planar structure including the stereostructures of **6** was similar to that of **5** except that the double bond between C-11 and C-12 in **5** was reduced in **6**; this was supported by the $^1H-^1H$ COSY correlations of H-10/H-11/H-12/H-13. The ROESY correlation of δ_H 1.72 (H-12) and δ_H 4.04 (H-14) suggested H-12 should be α-oriented, which is the same as that of **1**. Thus, compound **6** was assigned as striatoid F.

Compounds **1**−**6** were tested for their enhancement of NGF-induced neutrite outgrowth (neuritogenic activity) using rat pheochromocytoma (PC-12) cells as a model system of neuronal differentiation. The cells were incubated with compounds **1**−**6** at concentrations of 10−40 μM in combination with NGF (20 ng/mL) as a positive control.

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Table 1. ^{1}H and ^{13}C NMR Spectroscopic Data for Compounds 1−3

	1[a] (DMSO-d_6)		2[a] (DMSO-d_6)		3[b] (CD_3OD)	
no.	δ_C	δ_H (m, J in Hz)	δ_C	δ_H (m, J in Hz)	δ_C	δ_H (m, J in Hz)
1	77.5	3.76 m	77.5	3.74 t (7.1)	79.8	3.92 dd (7.6, 6.5)
2	37.4	2.03 m	37.6	2.05 dd (15.6, 6.2)	38.3	2.24 dd (15.7, 6.2)
		2.47 m		2.47 m		2.67 overlapped
3	134.5		135.1		137.5	
4	138.0		136.3		137.8	
5	46.4	2.21 brs	42.1	2.54 m	43.9	2.75 d (12.1)
6	40.7		41.1		42.3	
7	26.9	1.32 brd (13.8)	26.0	1.27 brd (12.8)	27.6	1.25 m
		1.65 m		1.62 m		1.78 overlapped
8	27.0	1.04 m	26.7	1.06 brd (12.8)	27.2	1.50 brd (13.1)
		1.49 m		1.67 m		1.80 overlapped
9	51.3		51.9		53.2	
10	24.8	1.65 m	29.0	2.29 m	30.6	2.46 dd (17.4, 6.5)
				2.54 m		2.64 overlapped
11	35.4	0.98 m	124.9	5.43 m	126.8	5.53 m
		1.99 m				
12	43.2	1.61 m	137.1		138.7	
13	45.7	2.10 dd (12.9, 8.1)	43.5	2.92 d (8.8)	44.7	3.08 m
14	92.4	3.84 overlapped	91.7	3.95 d (8.8)	93.7	4.21 d (8.8)
15	103.0	4.33 d (8.0)	99.1	5.58 s	103.2	5.64 s
16	18.8	0.83 s	17.3	0.86 s	17.8	1.02 s
17	24.0	1.00 s	23.9	0.97 s	24.3	1.11 s
18	26.6	2.62 m	26.6	2.70 m	28.2	2.83 m
19	20.9	0.89 d (6.7)	21.3	0.89 d (6.7)	21.4	1.01 d (6.6)
20	22.0	0.88 d (6.7)	22.3	0.89 d (6.7)	22.3	1.00 d (6.6)
21	55.1	3.34 s				
22						
1′	105.8	4.70, s	104.3	4.76 s	105.5	4.86 overlapped
2′	79.9		76.7		101.2	
3′	96.6		100.6		78.5	
4′	69.0	3.85 m	93.2		74.2	3.53 brs
5′α	63.6	3.52 dd (10.8, 5.0)	62.3	3.54 d (12.6)	61.8	3.99 d (13.6)
5′β		3.12 t (10.8)		3.28 d (12.6)		3.67 d (13.6)
1-OH		4.48 d (4.8)				
2′-OH		5.12 s				
3′-OH		5.86 s				
4′-OH		4.84 d (6.9)				
15-OH						

[a]^{1}H NMR 600 MHz, ^{13}C NMR 150 MHz. [b]^{1}HNMR 500 MHz, ^{13}C NMR 125 MHz.

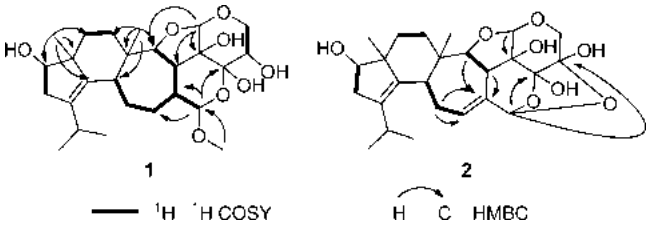

Figure 2. Key COSY and HMBC correlations of compounds **1** and **2**.

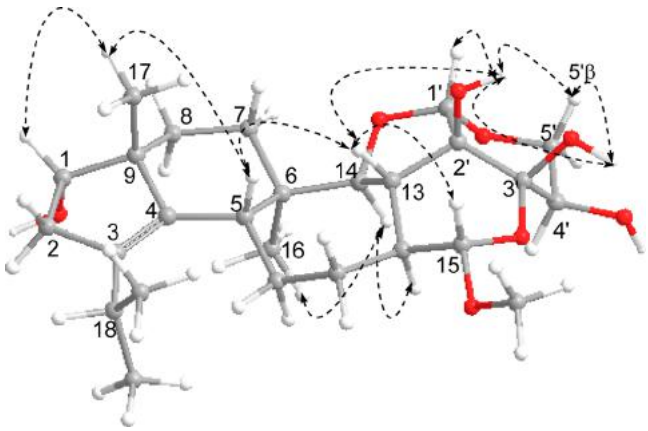

Figure 3. Key ROESY correlations of compound **1**.

Under these conditions, they all exerted a pronounced increase in neurite-bearing cells compared to control cells (Figures 5 and 6). In contrast, the compounds alone had no effect on neurite outgrowth in the absence of NGF.

In summary, six new metabolites, striatoids A−F (**1**−**6**), members of the cyathane diterpenoid-xyloside class of compounds, were isolated from cultures of the higher fungus *C. striatus*. Among them, both **2** and **3** carried a rare epoxy bridge between C-15 and C-4′. These metabolites were found to exert neurite outgrowth-promoting activity in NGF-mediated PC-12 cells.

■ EXPERIMENTAL SECTION

General Experimental Procedures. Optical rotations were recorded on an Autopol III automatic polarimeter (Rudolph Research Analytical). UV and IR spectra were obtained on a Thermo Scientific Evolution-300 UV−visible spectrophotometer and a Bruker Tensor 27 FT-IR spectrometer with KBr pellets. NMR spectra were obtained on Bruker Avance III 600 MHz and Bruker Avance III 500 spectrometers with tetramethylsilane as an internal standard at room temperature. High-resolution (HR) ESIMS were recorded on a Thermo Fisher Scientific Q-TOF mass spectrometer. Silica gel (300−400 mesh, Qingdao Marine Chemical Ltd., People's Republic of China) and RP-18 gel (20−45 μm, Fuji Silysia Chemical Ltd., Japan) were used for column chromatography (CC). Semipreparative HPLC was performed on a Waters 1100 liquid chromatography system equipped with a Hypersil BDS C_{18} column (4.6 mm × 250 mm; 10.0 mm × 250 mm). Fractions were monitored by TLC. Compounds were visualized by heating silica gel plates immersed in 10% H_2SO_4 in ethanol. Nutrient mixture F-12 (Ham) for the cell culture was purchased from Sigma-Aldrich (St. Louis, MO, USA). Horse serum (HS) was purchased from Gibco (Life Technologies Corporation, USA). Fetal bovine serum (FBS) was purchased from Hyclone (Thermo Scientific, China). The growth substrate poly-L-lysine was purchased from Sigma-Aldrich (St. Louis, MO, USA). Nerve growth factor was purchased from WuHan Hitech Biological Pharma Co., LTD, China.

Fungal Materials. The fungus *Cyathus striatus* (Huds. ex Pers.) Willd CFCC87405 was purchased from BNCL Biotechnology

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Table 2. ^{1}H and ^{13}C NMR Spectroscopic Data for Compounds 4−6

	4[a] (DMSO-d_6)		5[a] (DMSO-d_6)		6[b] (CD_3OD)	
no.	δ_C	δ_H (m, J in Hz)	δ_C	δ_H (m, J in Hz)	δ_C	δ_H (m, J in Hz)
1	77.0	3.72 overlapped	77.0	3.71 overlapped	80.0	3.95 m
2	37.7	2.06 dd (15.5, 5.4)	37.8	2.05 overlapped	38.0	2.24 m
		2.49 overlapped		2.50 overlapped		2.65 dd (15.8, 8.1)
3	135.0		135.2		136.5	
4	136.1		135.9		139.4	
5	42.4	2.72 d (12.7)	42.3	2.70 d (12.9)	48.6	2.36 d (8.7)
6	41.2		41.2		42.2	
7	26.0	1.22 m	26.1	1.21 brd (13.7)	27.8	1.22 brd (13.7)
		1.68 td (13.6, 3.8)		1.68 td (13.7, 3.8)		1.83 m
8	26.4	1.02 brd (13.6)	26.5	1.03 brd (13.7)	28.3	1.54 brd (13.7)
		1.76 td (13.6, 4.3)		1.76 td (13.7, 4.5)		1.66 td (13.7, 4.2)
9	52.0		52.1		52.7	
10	30.1	2.36 brd (17.4)	30.2	2.36 brd (17.0)	26.6	1.75 m
		2.55 overlapped		2.56 brt (17.0)		
11	121.6	5.44 dd (5.3, 2.4)	122.3	4.97 dd (5.2, 2.4)	36.7	1.07 m
						2.24 m
12	142.4		137.8		46.1	1.72 m
13	42.8	2.93 brs	42.6	2.97 brs	47.0	2.04 m
14	95.2	3.96 d (6.1)	95.2	3.96 d (6.1)	94.8	4.04 overlapped
15	70.1	4.47 brs	73.0	5.86 brs	86.2	3.85 overlapped
16	16.8	0.81 s	16.8	0.82 s	19.6	1.02 s
17	24.1	0.99 s	24.2	0.99 s	24.5	1.14 s
18	26.3	2.77 overlapped	26.4	2.76 m	28.2	2.78 m
19	21.3	0.92 d (6.5)	21.4	0.90 d (6.9)	21.1	1.01 d (6.7)
20	22.1	0.91d (6.5)	22.1	0.89 d (6.9)	22.4	1.00 d (6.7)
21			170.6		172.6	
22			21.1	2.07 s	20.7	2.11 s
1′	104.7	4.82 s	104.5	4.87 s	108.3	4.95 s
2′	78.3		78.3		81.2	
3′	74.1	3.69 overlapped	74.2	3.72 overlapped	81.6	4.11 overlapped
4′	73.4		72.6		71.8	
5′α	65.4	3.70 d (11.6)	65.5	3.67 d (11.7)	65.6	3.82 d (11.7)
5′β		3.04 d (11.6)		3.16 d (11.7)		3.18 d (11.7)
1-OH		4.43 d (4.9)		4.46 d (4.9)		
2′-OH		4.83 s		5.03 s		
3′-OH		5.28 d (3.9)		5.59 d (3.5)		
4′-OH		4.61 s		5.08 s		
15-OH		4.72 d (6.7)				

[a] ^{1}H NMR 600 MHz, ^{13}C NMR 150 MHz. [b] ^{1}HNMR 500 MHz, ^{13}C NMR 125 MHz.

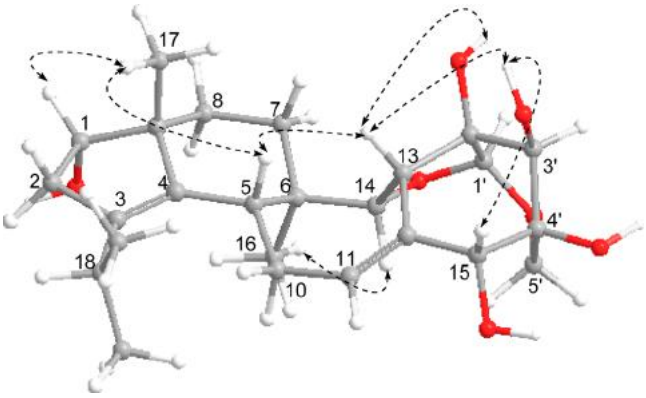

Figure 4. Key ROESY correlations of compound **4**.

Institute, Beijing. A specimen (No. HD20140303) was deposited at the College of Science, Northwest A&F University, Shaanxi, China. The culture medium consisted of glucose (1%), malt extract (1%), and yeast extract (0.4%), with pH 5.5. Fermentation was carried out on a shaker at 130 rpm for 22 days at 28 °C.

Extraction and Isolation. The culture broth (40 L) was filtered, and the filtrate was concentrated to 10 L, then extracted with ethyl acetate (10 L × 3), while the mycelium was extracted three times with $CHCl_3$−MeOH (1:1). The EtOAc layer together with the mycelium extract was concentrated under reduced pressure to give a crude extract (26.5 g), and the latter was applied to a silica gel column eluted with a gradient of petroleum−acetone (5:1 → 0:1) to obtain five fractions, A−E. Fraction C was separated by RP-18 (MeOH−H_2O, 10−100%) to give nine fractions (C1−C9). Fraction C7 was subjected to Sephadex LH-20 (MeOH) and silica gel CC (petroleum ether−EtOAc, 4:1) and further purified by semipreparative HPLC (70%, MeOH−H_2O, 10 mL/min) to yield **2** (t_R = 11 min, 6.0 mg) and **3** (t_R = 14 min, 8.3 mg). Fraction D was separated by RP-18 (10% → 100%, MeOH−H_2O) to obtain seven subfractions (D1−D7). D6 was purified by silica gel CC ($CHCl_3$−MeOH, 60:1) followed by Sephadex LH-20 (MeOH) to obtain two mixtures (D6A and D6B). D6A was purified by semipreparative HPLC (70%, MeOH−H_2O, 10 mL/min) to yield **6** (t_R = 8 min, 5.0 mg) and **1** (t_R = 12 min, 18.7 mg). Finally,

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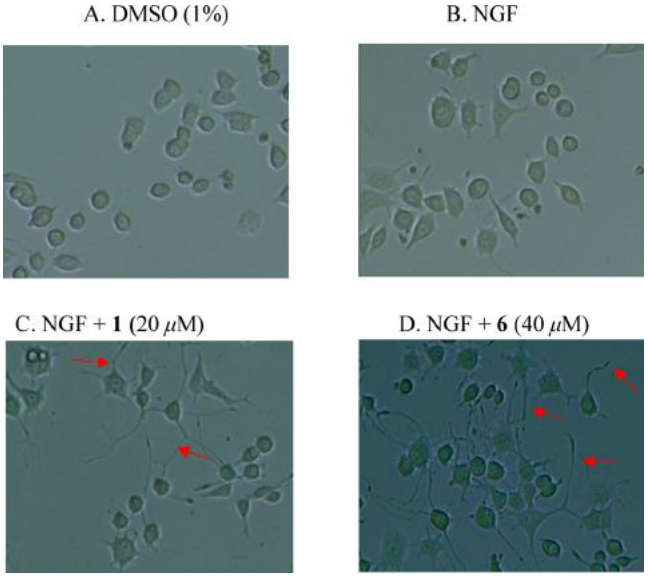

Figure 5. Effect of compounds **1** and **6** as examples on neurite outgrowth in PC-12 cells treated with 1% DMSO (A), NGF (20 ng/mL) (B), NGF (20 ng/mL) + **1** (20 μM) (C), and NGF (20 ng/mL) + **6** (40 μM) (D).

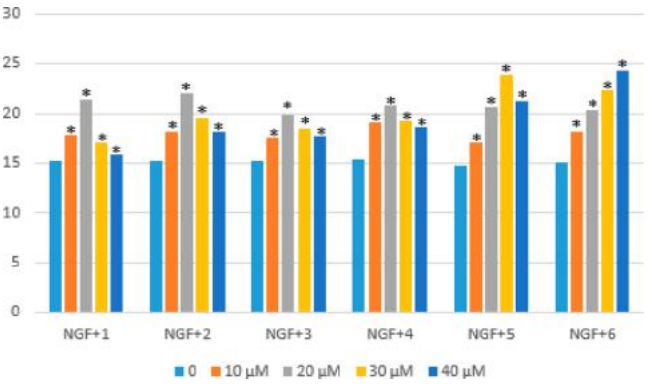

Figure 6. Effect of compounds **1**−**6** on neurite outgrowth in PC-12 cells ("0" NGF 20 ng/mL as positive control, $*p < 0.01$).

fraction D6B was applied to silica gel CC (petroleum ether−acetone, 6:1) to afford **4** (4.5 mg) and **5** (12.3 mg).

Striatoid A (1): white powder; $[\alpha]_D^{25}$ −58.4 (*c* 0.23, MeOH); IR (KBr) ν_{max} 3404, 2934, 2869, 1455, 1379, 1062, 1031 cm^{-1}; UV (MeOH) λ_{max} (log ε) 208 (3.86); ^1H (600 MHz) and ^{13}C NMR (150 MHz) data (DMSO-d_6), see Tables 1 and 2; positive ion HRESIMS *m/z* 503.2603 $[M + Na]^+$ (calcd for $C_{26}H_{40}O_8Na^+$, 503.2615).

Striatoid B (2): white powder; $[\alpha]_D^{25}$ −98.0 (*c* 0.20, MeOH); IR (KBr) ν_{max} 3412, 2959, 2930, 1631, 1075, 1028 cm^{-1}; UV (MeOH) λ_{max} (log ε) 208 (3.94); ^1H (600 MHz) and ^{13}C NMR (150 MHz) data (DMSO-d_6), see Tables 1 and 2; positive ion HRESIMS *m/z* 485.2139 $[M + Na]^+$ (calcd for $C_{25}H_{34}O_8Na^+$, 485.2146).

Striatoid C (3): white powder; $[\alpha]_D^{25}$ −66.2 (*c* 0.10, MeOH); IR (KBr) ν_{max} 3422, 2926, 2851, 1631, 1126, 1029 cm^{-1}; UV (MeOH) λ_{max} (log ε) 207 (3.75); ^1H (500 MHz) and ^{13}C NMR (125 MHz) data (CD_3OD), see Tables 1 and 2; positive ion HRESIMS *m/z* 447.2367 $[M + H]^+$ (calcd for $C_{25}H_{35}O_7$, 447.2377).

Striatoid D (4): white powder; $[\alpha]_D^{25}$ −137.9 (*c* 0.13, MeOH); IR (KBr) ν_{max} 3419, 2930, 1626, 1030 cm^{-1}; UV (MeOH) λ_{max} (log ε) 210 (3.78); ^1H (600 MHz) and ^{13}C NMR (150 MHz) data (DMSO-d_6), see Tables 1 and 2; positive ion HRESIMS *m/z* 471.2337 $[M + Na]^+$ (calcd for $C_{25}H_{36}O_7Na^+$, 471.2363).

Striatoid E (5): white powder; $[\alpha]_D^{25}$ −184.8 (*c* 0.16, MeOH); IR (KBr) ν_{max} 3417, 2954, 2837, 1726, 1031 cm^{-1}; UV (MeOH) λ_{max} (log ε) 210 (3.83); ^1H (600 MHz) and ^{13}C NMR (150 MHz) data (DMSO-d_6), see Tables 1 and 2; positive ion HRESIMS *m/z* 513.2444 $[M + Na]^+$ (calcd for $C_{27}H_{38}O_8Na^+$, 513.2482).

Striatoid F (6): white powder; $[\alpha]_D^{25}$ −54.4 (*c* 0.12, MeOH); IR (KBr) ν_{max} 3406, 2936, 1630, 1381, 1030 cm^{-1}; UV (MeOH) λ_{max} (log ε) 207 (3.65); ^1H (500 MHz) and ^{13}C NMR (125 MHz) data (CD_3OD), see Tables 1 and 2; positive ion HRESIMS *m/z* 515.2598 $[M + Na]^+$ (calcd for $C_{27}H_{40}O_8Na^+$, 515.2615).

Cell Culture. The rat adrenal pheochromocytoma cell line, PC-12, was obtained from China Center for Type Culture Collection. PC-12 cells were maintained in nutrient mixture F-12 (Ham) medium supplemented with 10% inactivated HS, 5% inactivated FBS, 100 U/mL penicillin G, 100 μg/mL streptomycin, and 2.5 g/L sodium bicarbonate at 37 °C in humidified air containing 5% CO_2.[11]

Analysis of Neurite Outgrowth in PC-12 Cells. Morphological analysis and quantification of neurite-bearing cells were perfromed using a phase-contrast microscope as described previously.[12] PC-12 cells were seeded on poly-L-lysine-coated 24-well plates at a density of 2×10^4 cells/mL in normal serum medium for 24 h. The F-12 medium containing low serum (1% HS and 0.5% FBS) was replaced prior to exposure to vehicle (0.1% DMSO) or indicated reagents. The cells were treated with tested compounds at various concentrations ranging from 10 to 40 μM with 20 ng/mL of NGF. Cells without treatment served as a negative control. Cells treated with 20 ng/mL of NGF served as a positive control. One concentration experiment was repeated in three wells. After an additional 48 h of incubation, neurite outgrowth of PC-12 cells was observed under an inverted microscope using phase-contrast objectives and photographed by the digital camera. Five images were selected randomly under a microscope for each well. At least 100 cells in each of five randomly separated fields were scored, and the cells with neurites greater than or equal to the length of one cell body were positive for neurite outgrowth and expressed as a percentage of the total cell number in five fields. Experiments were repeated at least three times, and data are expressed as mean ± SD ($**p < 0.01$).[12]

■ ASSOCIATED CONTENT

Supporting Information

NMR spectra of compounds **1**−**6**. This material is available free of charge via the Internet at http://pubs.acs.org.

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Notes

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